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Distributed **S**ystems

Activity 1.2.

A network file system

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1. Introduction: distributed file systems

A distributed system is a set of computers interconnected which share a state, offering a single system version.

By coordinating independent nodes, these architectures achieve key advantages such as horizontal scalability, high availability with fault tolerance, efficient parallel processing, lower hardware costs, and reduced latency through geographic distribution. To maintain this single-system view, they rely on fundamental properties including transparency, scalability, consistency, reliability and fault tolerance.

Keeping these properties in mind, we will analyze if the SZA/SAR service and the cloud system comply with them and what is required for them to be more complete.

2. SAR/SZA service

This application allows the user to access certain features depending on their level of authentication. Anonymous users can download files from the server, while authenticated users can also upload and delete files from the server. Additionally, there is no option to manage folders. The application uses the TCP protocol.

2.1 Properties

These are the properties of a distributed file system that fulfills the SZA service:

- **Permanent Information Storage**: A file system must provide permanent storage for information. In that case the SZA service satisfies this property because it is able to upload, delete and download files.
- **Security and access protection**: Implements the credential authentication because to login the service is necessary to enter password a username.
- **Concurrency**: SZA service uses TCP protocol, because of that the server can handle concurrent client connections.
- **File identifications**: The names of the files are identified by using ASCII character chains.

3. Cloud system

A cloud file storage system (such as Dropbox, EHUbox, or Google Drive) is an advanced evolution of distributed systems aimed at completely abstracting the distribution of the underlying hardware, software, and network infrastructure.

Unlike traditional tools for explicit file transfer, a cloud system operates transparently in the background, allowing users to work with their local file system as if there were no network involved.

3.1 Properties

Modern cloud file systems are built upon the fundamental properties of distributed systems and extend them with technical and user-oriented features. Cloud-specific technical properties:

- **Transparency (Identification and Location):** Users interact with files as if they were local, without needing to know their physical location on the servers.
- **Scalability and Fault tolerance:** The system manages the growth of users and data avoiding bottlenecks (through replication and caching), and survives client or server failures without compromising the service.
- **Consistency:** Maintains a coherent view of files across all concurrent devices, managing copies and replicas.
- **High availability and Offline operation:** The service remains operational despite partial failures and allows clients to work in disconnected mode, synchronizing data when the connection is restored.
- **Automatic synchronization:** Files are updated across multiple devices in the background, without requiring explicit uploads or downloads.
- **Versioning and Collaboration:** Support for a history of changes, recovery of deleted files (immutability/versions), and tools for sharing and co-editing.

4. Comparison between SAR/SZA service and cloud system

We are going to compare SZA service with the cloud system, taking into account all the properties of a distributed file system.

Distributed file system property	SAR/SZA Service	Cloud File System
Permanent information storage	Yes	Yes
File identification	Yes	Yes
Concurrent access	Yes	Yes
Security and access protection	Yes	Yes
Transparency	No. There is no directory hierarchy or management	Yes
Scalability	No. It relies on a single centralized server	Yes
Robustness to failures	No. Non-idempotent operations that cannot automatically recover	Yes
Consistency	No. It defines no client-side caching mechanism or concurrency control for simultaneous reads and writes to the same file	Yes
High availability and fault tolerance	No. It lacks server replication, load balancing, or local caching capabilities for disconnected operation	Yes

5. Conclusion

The SAR/SZA service provides a basic file transfer mechanism based on the client-server model, fulfilling the primary functions of persistent storage, authentication, and reliable data transfer over TCP. However, it lacks all forms of transparency (location, access, and presence), offline operation, and automatic synchronization that characterize a modern cloud system.